

DANG TRONG PHUC

Backend Developer · Game Server Engineering · Distributed Systems & Microservices
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PROFESSIONAL SUMMARY

Backend Developer with **3+ years** of experience specializing in high-concurrency game server engineering and scalable distributed systems. **Dual expertise in C++17** (real-time game server logic, sub-50ms latency) and **Java/Spring Boot** (microservices, event-driven architecture). Proven track record delivering production backends stress-tested at **10,000+ concurrent users**, achieving ~30% DB load reduction and ~25% latency improvement. Hands-on with gRPC/Protobuf, Kafka, RabbitMQ, Redis, and Docker across Agile cross-functional teams.

TECHNICAL SKILLS

Languages	C++ (C++11/14/17, STL, multithreading, memory management), Java (8, 21), JavaScript / TypeScript (basic)
Backend	Spring Boot 3, Spring Cloud, Spring Security, Spring Data JPA/Hibernate, RESTful API, gRPC, Protocol Buffers, WebSocket
Game Server	Real-time State Management, Round Lifecycle, Matchmaking Engine, Room/Session/Player State, Thread-Safe Concurrency, Race Condition Prevention, Authoritative Server Design, Socket Programming
Architecture	Microservices, Event-Driven Architecture, Distributed Systems, API Gateway (Spring Cloud Gateway), Service Discovery
Messaging & Async	Apache Kafka, RabbitMQ — async pipelines, pub/sub, event sourcing, background task processing
Database & Cache	MySQL, PostgreSQL, Redis (caching, session, pub/sub, leaderboard, rate-limiting)
DevOps / Tools	Docker, Linux (Ubuntu/CentOS), Git, Maven, Postman, CI/CD (basic)
Performance	SQL query optimization, Redis caching strategies, load testing (10k+ concurrent users), memory & CPU profiling, concurrency tuning

PROFESSIONAL EXPERIENCE

Gihot Studio

Ho Chi Minh City, Vietnam

Backend Developer — Game Server (C++ & Java)

Apr 2024 – Present

Tech: C++17, Java, Spring Boot, gRPC, Protocol Buffers, Redis, RabbitMQ, MySQL | Team: 20+ | Agile/Scrum

- Designed C++17 server-side game logic — round lifecycle, matchmaking engine, reward distribution, mission tracking — for real-time multiplayer mobile games supporting **5,000+ concurrent sessions**.
- Engineered thread-safe room/player state management using mutex locks, thread pools, and atomic operations; eliminated race conditions ensuring consistent gameplay under high-concurrency loads.
- Built gRPC services with Protobuf schemas (auth, matchmaking, in-game events, state sync) achieving **sub-50ms** client-server RTT; diagnosed and resolved complex sync bugs through systematic root cause analysis.
- Optimized Redis caching and session layers, reducing DB load by **~30%** and API response time by **~20%**; implemented RabbitMQ async pipelines for game event processing and background tasks.
- Stress-tested backend at **10,000+ concurrent virtual users**; identified and resolved bottlenecks achieving **25% end-to-end latency reduction** in production-like environments.
- Built Spring Boot microservices for GM Tool admin dashboard, game config management, and real-time monitoring; collaborated with client, QA, and DevOps teams across Agile sprints.

Hahalolo Travel Social Network

Ho Chi Minh City, Vietnam

Backend Developer — Java / Spring Boot

Jun 2022 – May 2023

Tech: Java 8, Spring Boot, Spring Security, JWT, JPA/Hibernate, MySQL, Docker, Linux | Team: 10+ | Agile

- Designed and built scalable RESTful APIs for a high-traffic social travel platform; ensured maintainability and reliable performance under production load.
- Implemented Spring Security + JWT authentication with role-based access control (RBAC), securing all API endpoints across the platform.
- Profiled and optimized SQL queries and JPA/Hibernate layers, reducing query execution time by **~40%** and improving overall system throughput.
- Developed real-time image processing features integrated into backend APIs, improving user interaction response speed by 20%.
- Collaborated with QA and DevOps on Docker-based deployments and Linux production incident debugging, contributing to system reliability improvements.

KEY PROJECTS

Dau Truong Huyen Thoai — Confidential MOBA/MMO Mobile Game (C++ Backend)

Feb 2025 – Present

Role: Backend Developer · Company: Gihot Studio
Tech: C++17, gRPC, Protocol Buffers, Redis, Event-Driven Architecture, Linux Server Environment

- Architected high-performance MOBA/MMO backend using C++17 and event-driven architecture; designed full Protobuf schema suite covering authentication, matchmaking, in-game events, economy, and hero/player stat systems.
- Built **Redis-centric data layer** for real-time session management, leaderboard systems, rate-limiting, and pub/sub messaging; developed thread-safe server logic with focus on CPU efficiency and lock contention minimization to support **horizontal scaling**.
- Performed performance profiling, latency benchmarking, and **stress testing**; built logging/monitoring observability stack — system fully prepared for live player traffic ramp-up.

Sieu Anh Hung Dai Chien — Real-Time Card Battle Mobile Game

Apr 2024 – Nov 2024

Role: Backend Developer · Company: Gihot Studio
Tech: Java 8, Spring Boot, Protobuf, Redis, MySQL, Socket

- Built microservices backend for real-time card game; reduced bottlenecks by **25%** and optimized for **5,000+ concurrent users** in load tests.
- Engineered high-concurrency socket layer with Protobuf serialization; implemented reward/mission/session systems with strict state consistency.
- Developed GM Tool REST APIs for live game configuration and real-time monitoring.

FPM — Financial Portfolio Manager

Oct 2025 – Present

Role: Backend Developer · Type: Personal Project (Cloud-Native Microservices)
Tech: Java 21, Spring Boot 3.5, Spring Cloud 2025.0, gRPC (Protobuf), Kafka, RabbitMQ, Redis 7, MySQL 8, Docker Compose, Android (Kotlin)

- Architected a **10-service cloud-native system** (user-auth, wallet, transaction, reporting, notification, OCR, AI, API Gateway, Eureka, Config Server) across 7 domains: Identity, Wallet, Transaction, Reporting, Notification, OCR, and AI-powered categorization.
- Designed **dual-broker event-driven architecture**: Kafka (8 topics) for high-throughput transaction streaming; RabbitMQ for domain event routing (wallet.created, balance.changed, budget.alerts) — fully decoupled async pipelines.
- Built **gRPC inter-service layer** with 6 Protobuf contracts; implemented Anti-Corruption Layer pattern; achieved sub-50ms internal RPC latency via HTTP/2 + binary Protobuf.
- Implemented **dual-layer JWT security**: Gateway validates token + Redis blacklist; downstream services re-validate via shared *fpm-security* library with row-level ownership checks (defense-in-depth vs. Mass Assignment attacks).
- Built **9 shared internal libraries** (fpm-bom, fpm-core, fpm-security, fpm-grpc, fpm-proto, fpm-domain, fpm-messaging, fpm-common, fpm-testing) to standardize cross-service response format, exception handling, JWT, and event publishing.
- Applied **resilience patterns**: Resilience4j Circuit Breaker (50% threshold, 30s open), Token Bucket rate limiting (5 req/5min login, 100 req/s standard), RabbitMQ retry with backoff; Redis multi-layer caching (TTL 5min) reducing DB load.

EDUCATION & LANGUAGES

Posts and Telecommunications Institute of Technology (PTIT)
[Bachelor of Engineering — Computer Electronic Engineering](#)
Ho Chi Minh City · Sep 2018 – Dec 2023

English	B1 – Aptis Certificate (ESOL); actively improving towards B2
Vietnamese	Native proficiency

CORE COMPETENCIES

High-Concurrency C++ · Real-Time Server Architecture · Authoritative Game Server · Matchmaking System · Game State Synchronization · Round Lifecycle Control · Thread-Safe Concurrency · gRPC / WebSocket · Redis Leaderboard & Session · Microservices Architecture · Event-Driven Architecture · Apache Kafka · RabbitMQ · Spring Boot · API Gateway · Spring Security + JWT · SQL Optimization · JPA/Hibernate · Docker · Linux · Load Testing · Performance Profiling · Production Debugging · Root Cause Analysis · Distributed Systems · Horizontal Scaling · High-Availability Systems · Agile / Scrum · Cross-functional Collaboration · System Design · Code Review · Technical Documentation

PROFESSIONAL STRENGTHS

Performance-first mindset — profile before optimizing, benchmarks over assumptions	Deep debugging skills — root cause analysis on complex production incidents	Cross-team collaboration — client, QA, DevOps — Agile sprint coordination
System design thinking — scalability and fault-tolerance from day one	Clean architecture — maintainable, well-documented, reviewable code	Proactive learner — C++17, Java 21, distributed systems in parallel